

Social Choice

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Outline

- 1 Introduction to Social Choice
- 2 Peer-Grading in MOOCs
 - Preliminaries
 - Correctness of Recovered Pairwise Rankings
 - Proofs



The Setting of Social Choice

Take voting scheme for example.

- A set O of **outcomes** (i.e., alternatives, candidates, etc.)
- A set A of agents s.t. each of them has a **preference** $\succ$ over the outcomes.
- The **social choice function**: a mapping from the profiles of the preferences to a particular outcome.



Outcomes & preferences

outcomes : a, b, c, d

$d \succ b \succ a \succ c$

preferences



d	b	a	a	a
b	c	b	c	b
a	a	c	b	c
c	d	d	d	d



Preferences

- A binary relation $\succsim$ such that
 - for every $a, b \in O$, $a \neq b$, we have either $a \succsim b$ or $b \succsim a$ but NOT both.
 - for $a, b, c \in O$, if $a \succsim b$ and $b \succsim c$, then we have $a \succsim c$.
- $\precsim$ can be defined similarly.
 - $\precsim: \neg \succsim$



Agents with preferences

- E.g., five agents (voters).
- Each agent has its preference over four candidates $\{a, b, c, d\}$.

outcomes : a, b, c, d

$d \succ b \succ a \succ c$

preferences



d	b	a	a	a
b	c	b	c	b
a	a	c	b	c
c	d	d	d	d

Agents with preferences

- E.g., three agents (voters).
- Each agent has its preference over four candidates $\{a, b, c, d\}$.

v_1	v_2	v_3
d	b	a
b	c	b
a	a	c
c	d	d

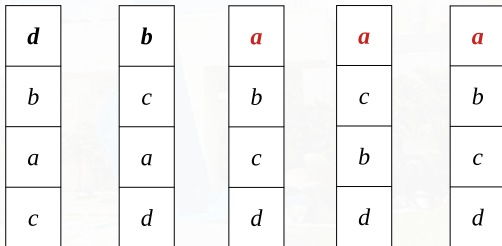


Plurality rule $\Rightarrow a$

outcomes : a, b, c, d

$d \succ b \succ a \succ c$

preferences



d	b	a	a	a
b	c	b	c	b
a	a	c	b	c
c	d	d	d	d

- Plurality rule: each agent can only give score 1 to the most preferred one and 0 to the others.



Plurality rule (contd.)

v_1	v_2	v_3
d	b	a
b	c	b
a	a	c
c	d	d

- Plurality rule: each agent can only give score 1 to the most preferred one and 0 to the others.



Plurality rule (contd.)

v_1	v_2	v_3
<i>d</i>	<i>b</i>	<i>a</i>
<i>b</i>	<i>c</i>	<i>b</i>
<i>a</i>	<i>a</i>	<i>c</i>
<i>c</i>	<i>d</i>	<i>d</i>

- Plurality rule:

Plurality rule (contd.)

v_1	v_2	v_3
<i>d</i>	<i>b</i>	<i>a</i>
<i>b</i>	<i>c</i>	<i>b</i>
<i>a</i>	<i>a</i>	<i>c</i>
<i>c</i>	<i>d</i>	<i>d</i>

- Plurality rule: depending on the tie-breaking rule.



Condorcet rule

v_1	v_2	v_3
d	b	a
b	c	b
a	a	c
c	d	d

- Condorcet rule:

- a vs. b
- a vs. c
- a vs. d



Condorcet rule

v_1	v_2	v_3
d	b	a
b	c	b
a	a	c
c	d	d

- Condorcet rule:

- a vs. $b \rightarrow b$
- a vs. $c \rightarrow a$
- a vs. $d \rightarrow a$



Condorcet rule

v_1	v_2	v_3
d	b	a
b	c	b
a	a	c
c	d	d

- Condorcet rule:

- c vs. a
- c vs. b
- c vs. d

Condorcet rule

v_1	v_2	v_3
d	b	a
b	c	b
a	a	c
c	d	d

- Condorcet rule:

- c vs. $a \rightarrow a$
- c vs. $b \rightarrow b$
- c vs. $d \rightarrow c$



Condorcet rule

v_1	v_2	v_3
d	b	a
b	c	b
a	a	c
c	d	d

- Condorcet rule:

- b vs. a
- b vs. c
- b vs. d

Condorcet rule

v_1	v_2	v_3
d	b	a
b	c	b
a	a	c
c	d	d

- Condorcet rule:

- b vs. $a \rightarrow b$
- b vs. $c \rightarrow b$
- b vs. $d \rightarrow b$



Condorcet rule

v_1	v_2	v_3
d	b	a
b	c	b
a	a	c
c	d	d

- Condorcet rule: b

- b vs. $a \rightarrow b$
- b vs. $c \rightarrow b$
- b vs. $d \rightarrow b$



Borda rule

v_1	v_2	v_3																								
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- Borda count rule:

Borda rule

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- Borda count rule:

- score of a : $1 + 1 + 3 = 5$.
- score of b : $2 + 3 + 2 = 7$.
- score of c : $0 + 2 + 1 = 3$.
- score of d : $3 + 0 + 0 = 3$.



Borda rule

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Inefficiency of Borda Count

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- Who is the winner by Borda counting?

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- Who is the winner by Borda counting? a : 6, b : 7, c : 2.

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- Who is the winner by Borda counting? a : 6, b : 7, c : 2.
- Condorcet principle follows?



Inefficiency of Borda Count

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- Who is the winner under the plurality rule?



Inefficiency of Borda Count

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- Condorcet principle follows? $a \succ b$, $a \succ c$.
- Who is the winner under the plurality rule? a .



Successive elimination

a sequence of pairwise majority votes determined by a fixed order of candidates.

v_1	v_2	v_3
b	a	c
d	b	a
c	d	b
a	c	d

- Successive elimination with ordering $a \rightarrow b \rightarrow c \rightarrow d$:



Successive elimination

v_1	v_2	v_3
b	a	c
d	b	a
c	d	b
a	c	d

- Successive elimination with ordering $a \rightarrow \cancel{b} \rightarrow c \rightarrow d$:

Successive elimination

v_1	v_2	v_3
b	a	c
d	b	a
c	d	b
a	c	d

- Successive elimination with ordering $\cancel{a} \rightarrow \cancel{b} \rightarrow c \rightarrow d$:

Successive elimination

v_1	v_2	v_3
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Successive elimination

v_1	v_2	v_3
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- Successive elimination with ordering $\cancel{a} \rightarrow \cancel{b} \rightarrow \cancel{c} \rightarrow d$: d



Successive elimination

v_1	v_2	v_3
b	a	c
d	b	a
c	d	b
a	c	d

- Successive elimination with ordering $\cancel{a} \rightarrow \cancel{b} \rightarrow \cancel{c} \rightarrow d$: d
 - The issue: all of the agents prefer b to d !



Successive elimination

v_1	v_2	v_3
b	a	c
d	b	a
c	d	b
a	c	d

- Successive elimination with ordering $a \rightarrow b \rightarrow c \rightarrow d$: d
- Successive elimination with ordering $a \rightarrow c \rightarrow b \rightarrow d$:



Successive elimination

v_1	v_2	v_3
b	a	c
d	b	a
c	d	b
a	c	d

- Successive elimination with ordering $a \rightarrow b \rightarrow c \rightarrow d$: d
- Successive elimination with ordering $a \rightarrow c \rightarrow b \rightarrow d$: b



Successive elimination (sensitive to the agenda order)

v_1	v_2	v_3
b	a	c
d	b	a
c	d	b
a	c	d

- Successive elimination with ordering $a \rightarrow b \rightarrow c \rightarrow d$: d
- Successive elimination with ordering $a \rightarrow c \rightarrow b \rightarrow d$: b
- Successive elimination with ordering $b \rightarrow c \rightarrow a \rightarrow d$:



Successive elimination (sensitive to the agenda order)

v_1	v_2	v_3
b	a	c
d	b	a
c	d	b
a	c	d

- Successive elimination with ordering $a \rightarrow b \rightarrow c \rightarrow d$: d
- Successive elimination with ordering $a \rightarrow c \rightarrow b \rightarrow d$: b
- Successive elimination with ordering $b \rightarrow c \rightarrow a \rightarrow d$: a



Condorcet Winner vs. Plurality

- Let's say we have 1,000 agents each of which has a preference over three candidates A, B, C .
 - 499 agents for $A \succ B \succ C$.
 - 3 agents for $B \succ C \succ A$.
 - 498 agents for $C \succ B \succ A$.
- Who is the Condorcet winner?



Condorcet Winner vs. Plurality

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- Who is the Condorcet winner? **B**.



Condorcet Winner vs. Plurality

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- Who is the Condorcet winner? **B**.
- Who is the winner under the plurality rule? **A**.



Exercise

On Borda Count & Condorcet

We have five voters with the following preferences (ordering) over the outcomes A, B, C , and D .

- $B \succ C \succ A \succ D$.
- $B \succ D \succ C \succ A$.
- $D \succ C \succ A \succ B$.
- $A \succ D \succ B \succ C$.
- $A \succ D \succ C \succ B$.

Who is the winner by the Borda Count rule?

Who is the Condorcet winner?



Let's consider a practical application in MOOCs.



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 - e.g., Coursera, EdX.



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- Outsourcing the grading task to the students.
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 - ▶ Ask each student to grade a SMALL number of her peers' assignments.
 - Then merge individual rankings into a global one.



Terminologies

- $\mathcal{A}$: universe of n elements (students).
- (n, k) -grading scheme:
a collection $\mathcal{B}$ of size- k subsets (**bundles**) of $\mathcal{A}$, such that each element of $\mathcal{A}$ belongs to exactly k subsets of $\mathcal{B}$.
- The **bundle graph**:
Represent the (n, k) -grading scheme with a bipartite graph.
- $\prec_b$: a ranking of the element b contains (partial order).



The aggregation rule

An aggregation rule:

profile of partial rankings $\rightarrow$ complete ranking of all elements.

- Borda:

SPRING FEAST 2016 BALLOT

a	LE BLE D'OR	金色三季 JINSE SANJI	5
b	CRYSTAL SPOON	水晶勺 CRYSTAL SPOON	3
c	Bei Yuan Restaurant	北园餐厅 BEI YUAN RESTAURANT	1
d	Tasty Steak	TASTY TASTY	2
e	Capricciosa	卡普里奇奥萨 CAPRICCIOSA	4

SPRING FEAST 2016 BALLOT

a	LE BLE D'OR	金色三季 JINSE SANJI	5
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SPRING FEAST 2016 BALLOT

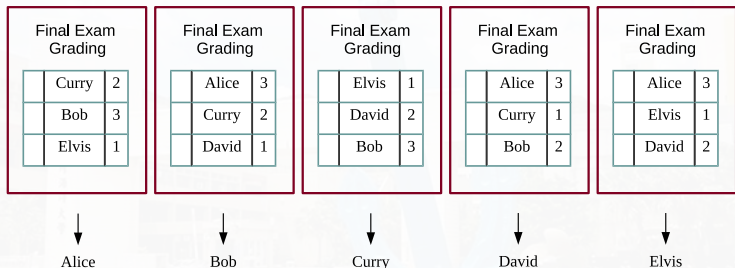
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- a: 14; b: 12; c: 4; d: 6; e: 9.

$a \succ b \succ e \succ d \succ c$.

Order-revealing grading scheme

An aggregation rule in peer grading (Borda):



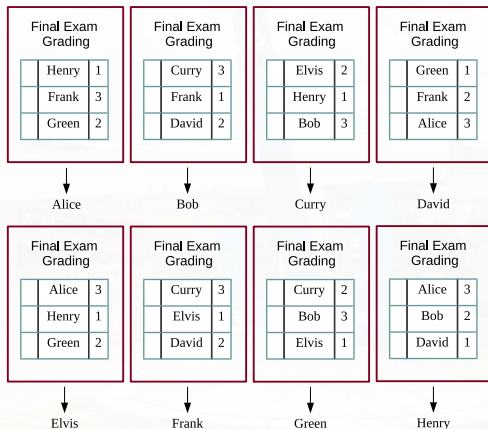
- Alice: 9; Bob: 8; Curry: 5; David: 5; Elvis: 3.

Alice $\prec$ Bob $\prec$ Curry $\prec$ David $\prec$ Elvis.

Assumption (perfect grading)

Each student grades the assignments in her bundle **consistently** to the ground truth.

Order-revealing grading scheme (contd.)

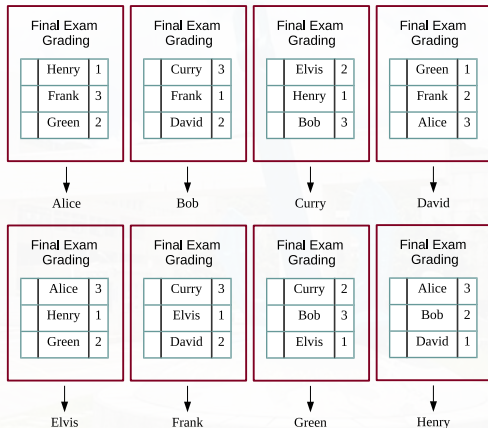


- Alice: 9; Bob: 8; Curry: 8; David: 5; Elvis: 4; Frank: 6; Green: 5; Henry: 3.

Alice $\prec$ Bob $\prec$ Curry $\prec$ Frank $\prec$ David $\prec$ Green $\prec$ Elvis $\prec$ Henry.



Order-revealing grading scheme (contd.)



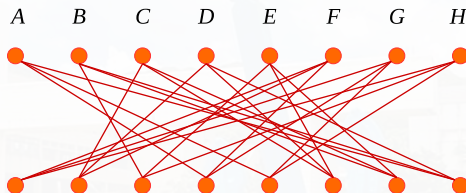
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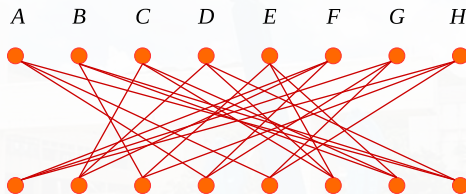
The bundle graph

The bundle graph:



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- A random k -regular graph:

A complete bipartite $K_{n,n} \rightarrow$ removing edges $\{v, v\}, \forall v \rightarrow$
repeat

“draw a perfect matching uniformly at random among all perfect matchings of the remaining graph”

for k times.



The limitation on the order revealing scheme

- The property of revealing the ground truth for certain:

$$\forall x, y \in \mathcal{A}, \exists B \in \mathcal{B} \text{ such that } x, y \in B.$$



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- Suppose NO bundle contains both $x, y \in \mathcal{A}$.
- Let $\prec, \prec'$ be two complete rankings.
 - x, y are in the first two positions in $\prec, \prec'$;
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 - Clearly, partial rankings within the bundles are identical in both cases.
 - No way to identify whether $\prec$ or $\prec'$ is the ground truth.
- To reveal the ground truth with certainty: $k = \Omega(\sqrt{n})$.
 - $n \cdot \binom{k}{2} \geq \binom{n}{2}$.



Seeking for approximate order-revealing grading schemes

- Use a bundle graph with a very low degree k (independent of n).
- Randomly permute the elements by $\pi : U \rightarrow \mathcal{A}$ before associating them to the nodes of U of the bundle graph.
- Aiming at $\frac{\text{\#correctly recovered pairwise relations}}{\binom{n}{2}}$.



The main result

Theorem (Caragiannis, Krimpas, Voudouris@AAMAS'15)

When

- Borda is applied as the aggregation rule, and
- all the partial rankings are consistent to the ground truth,

then the expected fraction of correctly recovered pairwise relations is $1 - O(1/\sqrt{k})$.



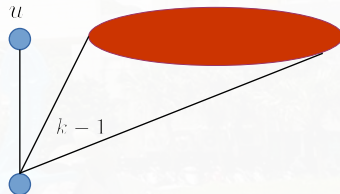
Question

- What will happen if we assign for each student only two assignments and each assignment is graded by exactly two students?



About who grading both u and v

- $\lambda_{u,v} := |N(u) \cap N(v)|$, for $u, v \in U$.
- $\sum_{v \in U \setminus \{u\}} \lambda_{u,v} = k(k-1)$.



- $W_{r,q}$: the r.v. denoting $B(a_r) - B(a_q)$ for $r < q$, $a_r, a_q \in \mathcal{A}$.
- $\Gamma_{u,v}^{r,q}$: the event that $\pi(u) = a_r$, $\pi(v) = a_q$.

$$C := \sum_{r=1}^{n-1} \sum_{q=r+1}^n \mathbb{E}[\mathbb{1}\{W_{r,q} > 0\}] = \sum_{r=1}^{n-1} \sum_{q=r+1}^n \Pr[W_{r,q} > 0]$$

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 &= \sum_{r=1}^{n-1} \sum_{q=r+1}^n \left(1 - \sum_{u,v \in U} \Pr[W_{r,q} \leq 0 \mid \Gamma_{u,v}^{r,q}] \cdot \Pr[\Gamma_{u,v}^{r,q}] \right)
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 \end{aligned}$$



- Given $\Gamma_{u,v}^{r,q}$,
 - the expected Borda score of a_r is $k + (k(k-1) - \lambda_{u,v}) \cdot \frac{n-r-1}{n-2} + \lambda_{u,v}$.
 - the expected Borda score of a_q is $k + (k(k-1) - \lambda_{u,v}) \cdot \frac{n-q}{n-2}$.
- Thus

$$\mathbb{E}[W_{r,q} \mid \Gamma_{u,v}^{r,q}] = (k(k-1) - \lambda_{u,v}) \frac{q-r-1}{n-2} + \lambda_{u,v}.$$



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Why?



Calculate the Borda score from another point of view

- Element a_r gets one point for each bundle it belongs to;
 - plus one additional point for each appearance of an element with rank higher $> r$ in the bundles a_r belongs to.



- In the bundles of containing a_r :
 - $\lambda_{u,v}$ appearances of a_q in the bundles of a_r .
 - $k(k-1) - \lambda_{u,v}$ appearances of elements different than a_r, a_q .
 - Each of them has prob. $\frac{n-r-1}{n-2}$ to have rank higher than r .



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- $\mathbb{E}[B(a_q) \mid \Gamma_{u,v}^{r,q}] = k + (k(k-1) - \lambda_{u,v}) \frac{n-q}{n-2}$.



Dealing with dependencies

Goal: $\Pr[W_{r,q} \leq 0 \mid \Gamma_{u,v}^{r,q}]$

- Given $\Gamma_{u,v}^{r,q}$, define $S = N(N(u) \cup N(v)) \setminus \{u, v\}$.
- $o : [|S|] \rightarrow S$ denotes an arbitrary ordering of nodes of S .
- X_i : the random variable denoting the rank of the element $\pi(o(i))$.
- Define the **Doob martingale** $Z_0, Z_1, \dots, Z_{|S|}$ such that
 - $Z_0 = \mathbb{E}[W_{r,q} \mid \Gamma_{u,v}^{r,q}];$
 - $Z_i = \mathbb{E}[W_{r,q} \mid \Gamma_{u,v}^{r,q}, X_1, \dots, X_i].$
- Hence, $W_{r,q} \mid \Gamma_{u,v}^{r,q} = Z_{|S|}$.



Martingale

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A sequence of random variables $Z_0, Z_1, \dots, Z_m$ is a **martingale** w.r.t. a sequence of random variables $X_1, X_2, \dots, X_m$ if $\forall i = 1, 2, \dots, m$,

$$\mathbb{E}[Z_i \mid X_1, X_2, \dots, X_{i-1}] = Z_{i-1}.$$

Doob martingale (Joseph L. Doob (1910–2004))

- W : a random variable
- $X_1, X_2, \dots, X_m$: a sequence of m random variables.

The sequence $Z_0, Z_1, \dots, Z_m$ such that

- $Z_0 = \mathbb{E}[W]$;
- $Z_i = \mathbb{E}[W \mid X_1, X_2, \dots, X_i], \forall i = 1, 2, \dots, m$

is called a **Doob martingale**.

Azuma-Hoeffding inequality

Azuma-Hoeffding inequality

Let $Z_0, Z_1, \dots, Z_m$ be a martingale with $Z_i - Z_{i-1} \leq c_i$ for $i = 1, \dots, m$. Then, for all $t > 0$,

$$\Pr[Z_m - Z_0 \leq -t] \leq \exp\left(-\frac{t^2}{2 \sum_{i=1}^m c_i^2}\right).$$



Dealing with dependencies (contd.)

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- Hence, $W_{r,q} \mid \Gamma_{u,v}^{r,q} = Z_{|S|}$.

Lemma 8

$\forall i \in \{1, 2, \dots, |S|\}$, it holds that $|Z_i - Z_{i-1}| \leq 2(\lambda_{u,o(i)} + \lambda_{v,o(i)})$.

Lemma 3

For every k -regular bipartite graph G ,

$$\theta_{u,v} = 4 \sum_{z \in N(N(u) \cup N(v)) \setminus \{u,v\}} (\lambda_{u,z} + \lambda_{v,z})^2 \leq 8k(k-1)(4k-3).$$

- Set $t = \mathbb{E}[W_{r,q} \mid \Gamma_{u,v}^{r,q}] (= Z_0)$, by the **Azuma-Hoeffding inequality**:

$$\begin{aligned}\Pr[Z_{|S|} - Z_0 \leq -t] &= \Pr[W_{r,q} \leq 0 \mid \Gamma_{u,v}^{r,q}] \\ &\leq \exp\left(-\frac{t^2}{2 \sum_{i=1}^m c_i^2}\right) \\ &= \exp\left(-\frac{\mathbb{E}[W_{r,q} \mid \Gamma_{u,v}^{r,q}]^2}{2\theta_{u,v}}\right).\end{aligned}$$



Back to the computation of C

$$\begin{aligned}
 C &= \sum_{r=1}^{n-1} \sum_{q=r+1}^n \left(1 - \frac{1}{n(n-1)} \sum_{u,v \in U} \Pr[W_{r,q} \leq 0 \mid \Gamma_{u,v}^{r,q}] \right) \\
 &\geq \sum_{r=1}^{n-1} \sum_{q=r+1}^n \left(1 - \frac{1}{n(n-1)} \sum_{u,v \in U} \exp \left(-\frac{E[W_{r,q} \mid \Gamma_{u,v}^{r,q}]^2}{2\theta_{u,v}} \right) \right) \\
 &= \sum_{r=1}^{n-1} \sum_{q=r+1}^n \left(1 - \frac{1}{n(n-1)} \sum_{u,v \in U} e^{-(\beta(u,v) \cdot y(q-r) + \delta(u,v))^2} \right) \\
 &= \frac{n(n-1)}{2} - \frac{1}{n(n-1)} \sum_{u,v \in U} \sum_{d=1}^{n-1} (n-d) e^{-(\beta(u,v) \cdot y(d) + \delta(u,v))^2} \\
 &\geq \frac{n(n-1)}{2} - \sum_{u,v \in U} \int_0^1 (1-y(d)) e^{-(\beta(u,v) \cdot y(d) + \delta(u,v))^2} dy
 \end{aligned}$$

- $\beta(u, v) = \frac{k(k-1) - \lambda_{u,v}}{\sqrt{2\theta_{u,v}}}$;
- $\delta(u, v) = \frac{\lambda_{u,v}}{\sqrt{2\theta_{u,v}}}$;
- $y(t) = \frac{t-1}{n-2}$.



$$\begin{aligned}
C &\geq \frac{n(n-1)}{2} - \sum_{u,v \in U} \int_0^1 (1-y) e^{-(\beta(u,v) \cdot y + \delta(u,v))^2} dy \\
&\geq \frac{n(n-1)}{2} - \sum_{u,v \in U} \frac{\beta(u,v) + \delta(u,v)}{2\beta(u,v)^2} \sqrt{\pi} \\
&\geq \frac{n(n-1)}{2} - \frac{k-1}{k(k-2)^2} \sqrt{\frac{\pi}{2}} \sum_{u,v \in U} \sqrt{\theta_{u,v}} \\
&\geq \frac{n(n-1)}{2} \left(1 - \frac{48\sqrt{2\pi}}{\sqrt{k}} \right).
\end{aligned}$$

Claim 9

Let $\beta > 0$, $\delta \geq 0$, then

$$\int_0^1 (1-y) e^{-(\beta y + \delta)^2} dy \leq \frac{\beta + \delta}{2\beta^2} \sqrt{\pi}.$$

- $k \geq 3$ (assumption).



Proof of Lemma 8

Recall:

- Given $\Gamma_{u,v}^{r,q}$, define $S = N(N(u) \cup N(v)) \setminus \{u, v\}$.
- $o : [|S|] \rightarrow S$ denotes an arbitrary ordering of nodes of S .
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- Hence, $W_{r,q} \mid \Gamma_{u,v}^{r,q} = Z_{|S|}$.

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$\forall i \in \{1, 2, \dots, |S|\}$, it holds that $|Z_i - Z_{i-1}| \leq 2(\lambda_{u,o(i)} + \lambda_{v,o(i)})$.

- $\mu_{u,v,w} = |N(u) \cap N(v) \cap N(w)|$.



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 - **increases** for each appearance of a_q in the same bundle with a_r ;
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 - **increases** for each appearance of $\pi(o(j))$ in a bundle containing BOTH a_r, a_q provided that $r < \text{rank}(\pi(o(j))) < q$;
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$$W_{r,q} = \lambda_{u,v}$$



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$$W_{r,q} = \lambda_{u,v} + \sum_{j=1}^{|S|} (\lambda_{u,o(j)} - \mu_{u,v,o(j)}) \cdot \mathbb{1}\{X_j > r\}$$



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 - **increases** for each appearance of a_q in the same bundle with a_r ;
 - **increases** for each appearance of $\pi(o(j))$ in a bundle containing a_r but NOT a_q provided that $r < \text{rank}(\pi(o(j))) < q$;
 - **increases** for each appearance of $\pi(o(j))$ in a bundle containing BOTH a_r, a_q provided that $r < \text{rank}(\pi(o(j))) < q$;
 - **decreases** for each appearance of $\pi(o(j))$ in a bundle containing a_q but NOT a_r provided that $\text{rank}(\pi(o(j))) > q$.

$$W_{r,q} = \lambda_{u,v} + \sum_{j=1}^{|S|} (\lambda_{u,o(j)} - \mu_{u,v,o(j)}) \cdot \mathbb{1}\{X_j > r\} + \sum_{j=1}^{|S|} \mu_{u,v,o(j)} \cdot \mathbb{1}\{r < X_j < q\}$$



Key to the proof of Lemma 8

- The Borda score difference $W_{r,q}$ (conditioned on $\Gamma_{u,v}^{r,q}$):
 - **increases** for each appearance of a_q in the same bundle with a_r ;
 - **increases** for each appearance of $\pi(o(j))$ in a bundle containing a_r but NOT a_q provided that $r < \text{rank}(\pi(o(j))) < q$;
 - **increases** for each appearance of $\pi(o(j))$ in a bundle containing BOTH a_r, a_q provided that $r < \text{rank}(\pi(o(j))) < q$;
 - **decreases** for each appearance of $\pi(o(j))$ in a bundle containing a_q but NOT a_r provided that $\text{rank}(\pi(o(j))) > q$.

$$\begin{aligned}
 W_{r,q} &= \lambda_{u,v} + \sum_{j=1}^{|S|} (\lambda_{u,o(j)} - \mu_{u,v,o(j)}) \cdot \mathbb{1}\{X_j > r\} + \sum_{j=1}^{|S|} \mu_{u,v,o(j)} \cdot \mathbb{1}\{r < X_j < q\} \\
 &\quad - \sum_{j=1}^{|S|} (\lambda_{v,o(j)} \cdot \mathbb{1}\{X_j > r\} - \lambda_{v,o(j)} \cdot \mathbb{1}\{X_j > q\})
 \end{aligned}$$



Key to the proof of Lemma 8

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$$\begin{aligned}
 W_{r,q} &= \lambda_{u,v} + \sum_{j=1}^{|S|} (\lambda_{u,o(j)} - \mu_{u,v,o(j)}) \cdot \mathbb{1}\{X_j > r\} + \sum_{j=1}^{|S|} \mu_{u,v,o(j)} \cdot \mathbb{1}\{r < X_j < q\} \\
 &\quad - \sum_{j=1}^{|S|} (\lambda_{v,o(j)} \cdot \mathbb{1}\{X_j > r\} - \lambda_{v,o(j)} \cdot \mathbb{1}\{X_j > q\}) \\
 &= \lambda_{u,v} + \sum_{j=1}^{|S|} (\lambda_{u,o(j)} \cdot \mathbb{1}\{X_j > r\} - \lambda_{v,o(j)} \cdot \mathbb{1}\{X_j > q\}).
 \end{aligned}$$



Key to the proof of Lemma 8 (contd.)

$$W_{r,q} = \lambda_{u,v} + \sum_{j=1}^{|S|} (\lambda_{u,o(j)} \cdot \mathbb{1}\{X_j > r\} - \lambda_{v,o(j)} \cdot \mathbb{1}\{X_j > q\})$$

$$\begin{aligned} Z_i - Z_{i-1} &= \mathbb{E}[W_{r,q} \mid \Gamma_{u,v}^{r,q}, X_1, X_2, \dots, X_i] - \mathbb{E}[W_{r,q} \mid \Gamma_{u,v}^{r,q}, X_1, X_2, \dots, X_{i-1}] \\ &= \sum_{j=i}^{|S|} \lambda_{u,o(j)} (\Pr[X_j > r \mid X_1, X_2, \dots, X_i] - \Pr[X_j > r \mid X_1, X_2, \dots, X_{i-1}]) - \\ &\quad \sum_{j=i}^{|S|} \lambda_{v,o(j)} (\Pr[X_j > q \mid X_1, X_2, \dots, X_i] - \Pr[X_j > q \mid X_1, X_2, \dots, X_{i-1}]) . \end{aligned}$$



Key to the proof of Lemma 8 (contd.)

$$\begin{aligned}
 Z_i - Z_{i-1} &= E[W_{r,q} \mid \Gamma_{u,v}^{r,q}, X_1, X_2, \dots, X_i] - E[W_{r,q} \mid \Gamma_{u,v}^{r,q}, X_1, X_2, \dots, X_{i-1}] \\
 &= \sum_{j=i}^{|S|} \lambda_{u,o(j)} (\Pr[X_j > r \mid X_1, X_2, \dots, X_i] - \Pr[X_j > r \mid X_1, X_2, \dots, X_{i-1}]) - \\
 &\quad \sum_{j=i}^{|S|} \lambda_{v,o(j)} (\Pr[X_j > q \mid X_1, X_2, \dots, X_i] - \Pr[X_j > q \mid X_1, X_2, \dots, X_{i-1}]) .
 \end{aligned}$$

Note that:

$$\Pr[X_j > r \mid X_1, X_2, \dots, X_{i-1}] = \frac{x+y}{n-i-1}, \quad \Pr[X_j > q \mid X_1, X_2, \dots, X_{i-1}] = \frac{y}{n-i-1} \quad (i \leq j \leq |S|)$$

$$\Pr[X_j > r \mid X_1, X_2, \dots, X_i] = \frac{x+y-\mathbb{1}\{X_i > r\}}{n-i-2}, \quad \Pr[X_j > q \mid X_1, X_2, \dots, X_i] = \frac{y-\mathbb{1}\{X_i > q\}}{n-i-2} \quad (i+1 \leq j \leq |S|)$$

- x : # available ranks from $[n] \setminus \{r, q, X_1, X_2, \dots, X_{i-1}\}$ that are between r and q ;
- y : # available ranks from $[n] \setminus \{r, q, X_1, X_2, \dots, X_{i-1}\}$ that are $> q$.



Key to the proof of Lemma 8 (contd.)

$$\begin{aligned}
 Z_i - Z_{i-1} &= \lambda_{u,o(i)} \left(\mathbb{1}\{X_i > r\} - \frac{x+y}{n-i-1} \right) - \lambda_{v,o(j)} \left(\mathbb{1}\{X_i > q\} - \frac{y}{n-i-1} \right) \\
 &+ \sum_{j=i+1}^{|S|} \lambda_{u,o(j)} \left(\frac{x+y - \mathbb{1}\{X_i > r\}}{n-i-2} - \frac{x+y}{n-i-1} \right) \\
 &+ \sum_{j=i+1}^{|S|} \lambda_{v,o(j)} \left(\frac{y - \mathbb{1}\{X_i > q\}}{n-i-2} - \frac{y}{n-i-1} \right)
 \end{aligned}$$

Note that:

$$\Pr[X_j > r \mid X_1, \dots, X_{i-1}] = \frac{x+y}{n-i-1}, \quad \Pr[X_j > q \mid X_1, \dots, X_{i-1}] = \frac{y}{n-i-1} \quad (i \leq j \leq |S|)$$

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- y : # available ranks from $[n] \setminus \{r, q, X_1, \dots, X_{i-1}\}$ that are $> q$.



Key to the proof of Lemma 8 (contd.)

$$\begin{aligned}
 Z_i - Z_{i-1} &= \lambda_{u,o(i)} \left(\mathbb{1}\{X_i > r\} - \frac{x+y}{n-i-1} \right) - \lambda_{v,o(j)} \left(\mathbb{1}\{X_i > q\} - \frac{y}{n-i-1} \right) \\
 &+ \sum_{j=i+1}^{|S|} \lambda_{u,o(j)} \left(\frac{x+y - \mathbb{1}\{X_i > r\}}{n-i-2} - \frac{x+y}{n-i-1} \right) \\
 &+ \sum_{j=i+1}^{|S|} \lambda_{v,o(j)} \left(\frac{y - \mathbb{1}\{X_i > q\}}{n-i-2} - \frac{y}{n-i-1} \right)
 \end{aligned}$$

Note that:

$$\Pr[X_j > r \mid X_1, \dots, X_{i-1}] = \frac{x+y}{n-i-1}, \quad \Pr[X_j > q \mid X_1, \dots, X_{i-1}] = \frac{y}{n-i-1} \quad (i \leq j \leq |S|)$$

$$\Pr[X_j > r \mid X_1, \dots, X_i] = \frac{x+y - \mathbb{1}\{X_i > r\}}{n-i-2}, \quad \Pr[X_j > q \mid X_1, \dots, X_i] = \frac{y - \mathbb{1}\{X_i > q\}}{n-i-2} \quad (i+1 \leq j \leq |S|)$$

- x : # available ranks from $[n] \setminus \{r, q, X_1, \dots, X_{i-1}\}$ that are between r and q ;
- y : # available ranks from $[n] \setminus \{r, q, X_1, \dots, X_{i-1}\}$ that are $> q$.



Key to the proof of Lemma 8 (contd.)

(Assume $n \geq 3k(k-1) + 2$; Note: $|S| \leq 2k(k-1)$)

$$\begin{aligned}
 Z_i - Z_{i-1} &= \lambda_{u,o(i)} \left(\mathbb{1}\{X_i > r\} - \frac{x+y}{n-i-1} \right) - \lambda_{v,o(j)} \left(\mathbb{1}\{X_i > q\} - \frac{y}{n-i-1} \right) \\
 &+ \sum_{j=i+1}^{|S|} \lambda_{u,o(j)} \left(\frac{x+y - \mathbb{1}\{X_i > r\}}{n-i-2} - \frac{x+y}{n-i-1} \right) \\
 &+ \sum_{j=i+1}^{|S|} \lambda_{v,o(j)} \left(\frac{y - \mathbb{1}\{X_i > q\}}{n-i-2} - \frac{y}{n-i-1} \right) \\
 &= \left(\lambda_{u,o(i)} - \frac{\sum_{j=i+1}^{|S|} \lambda_{u,o(j)}}{n-i-2} \right) \cdot \left(\mathbb{1}\{X_i > r\} - \frac{x+y}{n-i-1} \right) \\
 &+ \left(\lambda_{v,o(i)} - \frac{\sum_{j=i+1}^{|S|} \lambda_{v,o(j)}}{n-i-2} \right) \cdot \left(\frac{y}{n-i-1} - \mathbb{1}\{X_i > q\} \right).
 \end{aligned}$$

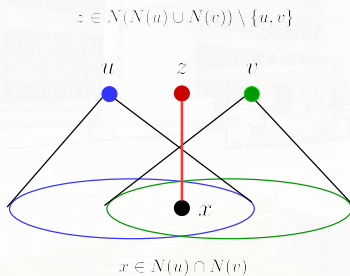


Key to the proof of Lemma 3

Lemma 3

For every k -regular bipartite graph G ,

$$\theta_{u,v} = 4 \sum_{z \in N(N(u) \cup N(v)) \setminus \{u,v\}} (\lambda_{u,z} + \lambda_{v,z})^2 \leq 8k(k-1)(4k-3).$$



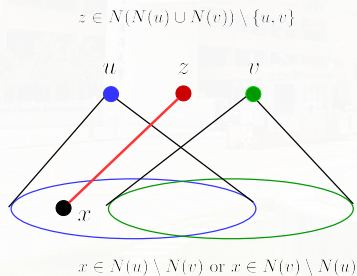
- The edge contributes 2 to the quantity $\lambda_{u,z} + \lambda_{v,z}$.
 - $\lambda_{u,z} + \lambda_{v,z} \leq 2k$.
- The edge contributes $\leq (2k)^2 - (2k-2)^2 = 8k-4$ to the quantity $(\lambda_{u,z} + \lambda_{v,z})^2$.
- There are $|N(u) \cap N(v)|(k-2)$ such edges.

Key to the proof of Lemma 3 (contd.)

Lemma 3

For every k -regular bipartite graph G ,

$$\theta_{u,v} = 4 \sum_{z \in N(N(u) \cup N(v)) \setminus \{u,v\}} (\lambda_{u,z} + \lambda_{v,z})^2 \leq 8k(k-1)(4k-3).$$



- The edge contributes 1 to the quantity $\lambda_{u,z} + \lambda_{v,z}$.
 - $\lambda_{u,z} + \lambda_{v,z} \leq 2k - 1$.
- The edge contributes $\leq (2k - 1)^2 - (2k - 2)^2 = 4k - 3$ to the quantity $(\lambda_{u,z} + \lambda_{v,z})^2$.
- There are $2(k - |N(u) \cap N(v)|)(k - 1)$ such edges.



Proof of the Gaussian integral (Claim 9)



$$\int_0^1 (1-y)e^{-(\beta y+\delta)^2} dy = \underbrace{\int_0^1 e^{-(\beta y+\delta)^2} dy}_A - \underbrace{\int_0^1 ye^{-(\beta y+\delta)^2} dy}_B$$

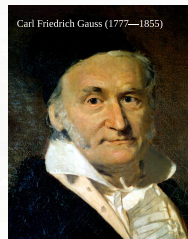
- The **error function**:

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt.$$

- The **Gaussian integral**:

$$\int_{-\infty}^{\infty} e^{-t^2} dt = \sqrt{\pi}.$$

$$\therefore \int_0^{\infty} e^{-t^2} dt = \frac{\sqrt{\pi}}{2} \quad \text{and} \quad \text{erf}(x) \leq \frac{2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = 1.$$



- $B = \int_0^1 ye^{-(\beta y + \delta)^2} dy.$

$$\text{Let } v = \beta y + \delta \quad \therefore \begin{cases} dv = \beta dy \\ y = \frac{v - \delta}{\beta} \end{cases}.$$

$$\begin{aligned} \therefore B &= \frac{1}{\beta} \int_{\delta}^{\beta + \delta} \frac{v - \delta}{\beta} \cdot e^{-v^2} dv \\ &= \frac{1}{\beta^2} \left(\int_{\delta}^{\beta + \delta} v \cdot e^{-v^2} dv - \delta \cdot \int_{\delta}^{\beta + \delta} e^{-v^2} dv \right) \\ &= \frac{1}{2\beta^2} \int_{\delta}^{\beta + \delta} e^{-v^2} dv^2 - \frac{\delta}{\beta^2} \cdot \frac{\sqrt{\pi}}{2} \cdot \frac{2}{\sqrt{\pi}} \int_{\delta}^{\beta + \delta} e^{-v^2} dv \\ &= \frac{1}{2\beta^2} \left(-e^{-(\beta + \delta)^2} + e^{-\delta^2} \right) - \frac{\delta\sqrt{\pi}}{2\beta^2} (\text{erf}(\beta + \delta) - \text{erf}(\delta)). \end{aligned}$$

- $A = \int_0^1 e^{-(\beta y + \delta)^2} dy.$

Let $u = \beta y + \delta \quad \therefore du = \beta dy.$

$$\begin{aligned} \therefore A &= \frac{1}{\beta} \int_{\delta}^{\beta + \delta} e^{-u^2} du \\ &= \frac{\sqrt{\pi}}{2\beta} \cdot \frac{2}{\sqrt{\pi}} \left(\int_0^{\beta + \delta} e^{-u^2} du - \int_0^{\delta} e^{-u^2} du \right) \\ &= \frac{\sqrt{\pi}}{2\beta} (\text{erf}(\beta + \delta) - \text{erf}(\delta)). \end{aligned}$$

- Thus

$$\begin{aligned} A - B &= \frac{\beta + \delta}{2\beta^2} \sqrt{\pi} (\text{erf}(\beta + \delta) - \text{erf}(\delta)) + \frac{1}{2\beta^2} (e^{-(\beta + \delta)^2} - e^{-\delta^2}) \\ &\leq \frac{\beta + \delta}{2\beta^2} \sqrt{\pi}. \end{aligned}$$

